CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A piece of Zn is placed in solutions of $FeSO_4$, $CuSO_4$ and $MgSO_4$. Reaction will occur in: 1

(a) all three **(b)** $MgSO_4$ only

(c) $FeSO_4$ and $CuSO_4$ **(d)** $CuSO_4$ only

2. The volume of CO_2 at STP on burning 1 mol ethane completely is: 1

(a) 89.6 L **(b)** 67.2 L **(c)** 22.4 L **(d)** 44.8 L

3. On adding dilute HCl to a solid X, a colourless gas evolves that turns lime water milky. X is likely: 1

(a) Na_2CO_3 **(b)** $NaCl$ **(c)** $NaNO_3$ **(d)** Na_2SO_4

4. A compound on heating gives CO_2; its aqueous solution is mildly basic and is used in baking. It is: 1

(a) NaCl **(b)** $NaHCO_3$ **(c)** Na_2CO_3 **(d)** NaOH

5. A black powder X when heated with carbon gives brown metal Y which on heating in air returns to X. Y is: 1

(a) iron **(b)** mercury **(c)** copper **(d)** sodium

6. Compound Z on heating with conc. H_2SO_4 gives ethene and on oxidation gives ethanoic acid. Z is: 1

(a) ethanol **(b)** methanol **(c)** ethane **(d)** ethanal

7. Assertion (A): A balanced chemical equation obeys the law of conservation of mass. Reason (R): The number of atoms of each element is equal on both sides. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. Assertion (A): Sodium hydrogen carbonate is used as an antacid. Reason (R): It reacts with HCl in the stomach to release CO_2 and neutralise excess acid. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. Assertion (A): Metals are good conductors of electricity. Reason (R): Metals have free electrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): Soap does not form lather in hard water. Reason (R): Hard water contains calcium and magnesium ions which form insoluble salts with soap. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. In the electrolysis of water, 20 mL of oxygen is collected. The volume of hydrogen collected is: 1

(a) 20 mL **(b)** 60 mL **(c)** 40 mL **(d)** 10 mL

12. If H^+ concentration of a solution is 10^{-3} M, its pH is: 1

(a) -3 **(b)** 3 **(c)** 11 **(d)** 7

13. A metal oxide X dissolves in water to turn red litmus blue. X is most likely: 1

(a) Fe_2O_3 **(b)** Al_2O_3 **(c)** Na_2O **(d)** CuO

14. 46 g of ethanol (C2H5OH) is completely burnt. The mass of CO_2 formed is: 1

(a) 88 g **(b)** 46 g **(c)** 44 g **(d)** 132 g

15. An iron nail is kept in CuSO4 solution for a while. Its mass: 1

(a) remains same **(b)** decreases
(c) increases slightly **(d)** becomes zero

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does an aluminium vessel not corrode easily though aluminium is above zinc in the reactivity series? 2

17. How many moles of H_2 are needed to saturate 1 mol of propyne? 2

18. Why does an iron nail kept for some time in copper sulphate solution show an increase in mass? 2

19. A solution of pH 3 is diluted 100 times. Predict its pH and state its acidity. 2

20. Why is the reaction of sodium with water not carried out in the lab with large pieces? Give the equation. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A student placed metals A, B, C, D in $FeSO_4$, $CuSO_4$ solutions. A and B displaced Fe and Cu; C displaced Cu only; D displaced none. Arrange in order of reactivity. 3

22. Calculate the volume of oxygen at STP needed to burn 2 mol of methane and the volume of CO_2 produced. 3

23. Element X is burnt in air with a dazzling white flame giving a white powder Y. Y dissolved in water turns red litmus blue. Identify X and Y and give the equation. 3

24. Explain the use of a pH meter or universal indicator to differentiate between a strong and weak acid of the same concentration. 3

OR

A black powder X on heating with hydrogen gives a brown metal Y. Y on heating in air gives X again. Identify and write equations. 3

25. Explain why graphite conducts but diamond does not, and why fullerene is a molecular solid. 3

OR

Copper powder is heated in a china dish and then hydrogen is passed over the product. Describe the changes and equations. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Zinc and copper are separately treated with dilute sulphuric acid. State the observations and equations, and explain using reactivity. 5

OR

A student mixes 25 mL of 0.1 M NaOH with 25 mL of 0.1 M HCl and 25 mL of 0.1 M NaOH with 50 mL of 0.1 M HCl. Predict pH regions and explain. 5

27. Explain the extraction of aluminium from bauxite: steps, equations at electrodes, and why cryolite is added. 5

OR

Calculate: 4.6 g of ethanol burns completely (C = 12, H = 1, O = 16). Find the mass of CO_2 and water, and volume of CO_2 at STP. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND